

Energy Boundedness via $Q(t)$ for Global Regularity

1 Energy Boundedness via $Q(t)$

One of the critical components for proving global regularity of the Navier–Stokes equations is the boundedness of the energy functional $Q(t)$, defined as:

$$Q(t) = \|u(t)\|_{L^2}^2 + \|\nabla u(t)\|_{L^2}^2. \quad (1)$$

This functional tracks the kinetic energy and enstrophy of the flow over time, ensuring that both remain finite and under control for all $t \geq 0$.

1.1 Energy Evolution and Dissipation

The time evolution of $Q(t)$ is governed by the energy equation:

$$\frac{dQ}{dt} = -2\nu\|\nabla u\|_{L^2}^2 - 2\nu\|\Delta u\|_{L^2}^2 + 2 \int_{\mathbb{R}^3} f \cdot u \, dx, \quad (2)$$

where f represents the external forcing term, and ν is the kinematic viscosity.

Dissipation Dominance: The dissipation terms $-2\nu\|\nabla u\|_{L^2}^2$ and $-2\nu\|\Delta u\|_{L^2}^2$ ensure that energy is dissipated effectively, counteracting nonlinear growth contributions from the convective term $(u \cdot \nabla)u$.

External Forcing Contribution: The term $2 \int f \cdot u \, dx$ represents energy input from external forces. Assuming $f \in L^2$, we bound this term as:

$$\left| \int_{\mathbb{R}^3} f \cdot u \, dx \right| \leq \|f\|_{L^2} \|u\|_{L^2}. \quad (3)$$

This ensures that energy growth due to external forcing is finite.

1.2 Gronwalls Inequality and Global Boundedness

Integrating Eq. (2) over time and applying Gronwalls inequality yields:

$$Q(t) \leq Q(0)e^{-\nu t} + \int_0^t \|f(s)\|_{L^2}^2 \, ds, \quad (4)$$

where $Q(0) = \|u_0\|_{L^2}^2 + \|\nabla u_0\|_{L^2}^2$ is the initial energy and enstrophy.

1.3 Numerical Validation

To verify the boundedness of $Q(t)$ numerically, consider the following setup:

Initial Data: Choose initial conditions $u_0 \in H^1$ such that $Q(0)$ is finite:

$$Q(0) = \|u_0\|_{L^2}^2 + \|\nabla u_0\|_{L^2}^2 < \infty. \quad (5)$$

Ensure that u_0 satisfies periodic boundary conditions or is compactly supported for simplicity in computational simulations.

Forcing Term: Select a forcing term $f \in L^2(\mathbb{R}^3)$ to model realistic energy input. Examples include:

- ****Time-independent forcing:**** $f(x) = f_0(x)$, where f_0 represents a stationary external field.
- ****Time-dependent forcing:**** $f(x, t) = f_0(x)e^{-\lambda t}$ to simulate decaying energy input.

Metrics for Validation: Track the following metrics over time to verify energy boundedness:

1. ****Functional Evolution:**** Compute $Q(t)$ at discrete time steps and compare its evolution to the theoretical bound in Eq. (4).
2. ****Dissipation vs. Forcing:**** Compare the contributions of dissipation ($-\nu\|\Delta u\|_{L^2}^2$) and forcing ($\|f\|_{L^2}\|u\|_{L^2}$) to ensure dissipation dominance.
3. ****Spectral Energy Decay:**** Evaluate the energy spectrum $E(k, t)$ and confirm spectral decay:

$$E(k, t) \sim k^{-\beta}, \quad \beta > 3, \quad (6)$$

for high wave numbers k .

Computational Framework: Simulate the Navier–Stokes equations using Fourier spectral methods or finite-element methods. Key steps include:

- ****Discretization:**** Use a pseudospectral scheme for periodic domains or finite elements for bounded domains.
- ****Time Stepping:**** Implement an implicit-explicit (IMEX) scheme to handle viscous and nonlinear terms efficiently.
- ****Resolution:**** Ensure sufficient grid resolution to capture small-scale dynamics, satisfying the condition:

$$\Delta x < \frac{1}{k_{\max}}, \quad (7)$$

where $k_{\max}$ is the maximum resolved wave number.

Validation Outcomes: Numerical validation should confirm:

- The boundedness of $Q(t)$ for all $t \geq 0$.
- Dominance of dissipation over forcing in the energy balance.
- Spectral decay consistent with theoretical predictions.

Conclusion

The functional $Q(t)$ provides a robust tool for ensuring energy boundedness and controlling enstrophy in the Navier–Stokes equations. Numerical validation offers empirical confirmation of its role in spectral decay, compactness, and gradient control, reinforcing its importance in proving global regularity.

References

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